SOLUTIONS: Feedback Systems

1. a) The transfer function of the circuit of Figure 1.2 is

$$\frac{Z_f(s)}{Z_i(s)} = -\frac{C_i(s+1/R_iC_i)}{C_f(s+1/R_fC_f)} = -\frac{C_is+1/R_i}{C_fs+1/R_f}$$

Substituting the values: $\boxed{G(s) = G_1(s)G_2(s) = \frac{1}{s(s+1)}}$. [5]

b) Since $y(s) = G(s)u(s)$ the differential equation is $\boxed{\ddot{y}(t)+\dot{y}(t) = u(t)}$. [5]

c) The Nyquist diagram is displayed on the next page. [5]

d) Let $x_1(t) = y(t), x_2(t) = \dot{y}(t)$ and $x(t) = \left[\begin{array}{cc} x_1(t) & x_2(t) \end{array}\right]^T$. So,

$$\boxed{\dot{x}(t) = \left[\begin{array}{cc} 0 & 1 \\ 0 & -1 \end{array}\right] x(t) + \left[\begin{array}{c} 0 \\ 1 \end{array}\right] u(t), \quad y(t) = \left[\begin{array}{cc} 1 & 0 \end{array}\right] x(t)} \qquad [5]$$

e) Substituting the state feedback law into the state equation gives

$$\dot{x}(t) = (A - B\left[\begin{array}{cc} k_1 & k_2 \end{array}\right])x(t) = \left[\begin{array}{cc} 0 & 1 \\ -k_1 & -(k_2+1) \end{array}\right] x(t).$$

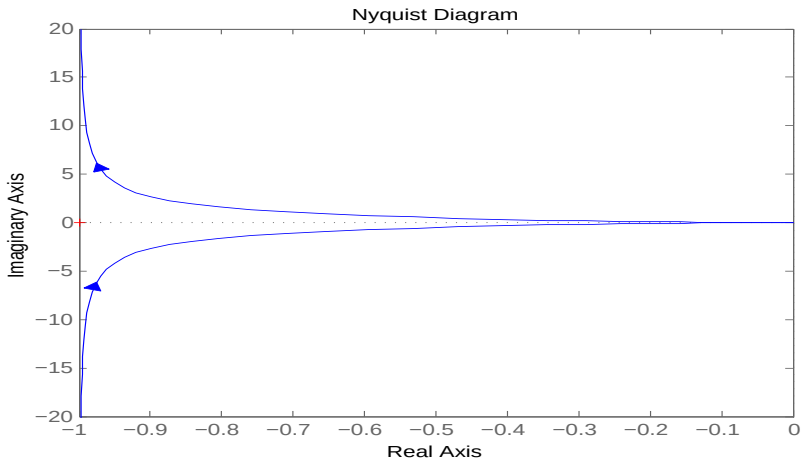
So the closed loop has a characteristic equation $s^2 + (1+k_2)s + k_1$. For pole locations at -1 and -2, this must be $s^2 + 3s + 2$ and so $\boxed{k_1 = k_2 = 2}$. [5]

f) Since $G(s)$ is type 1 and $\lim_{t\to\infty} r(t) = 1$, the steady state error is: $\boxed{e_{ss} = 0.}$ [5]

g) The closed–loop transfer function is $H(s) = \frac{K}{s^2+s+K}$. To satisfy the specifications, the closed–loop poles must be at $-0.5 \pm j0.5$. Thus $\boxed{K = 0.5.}$ [5]

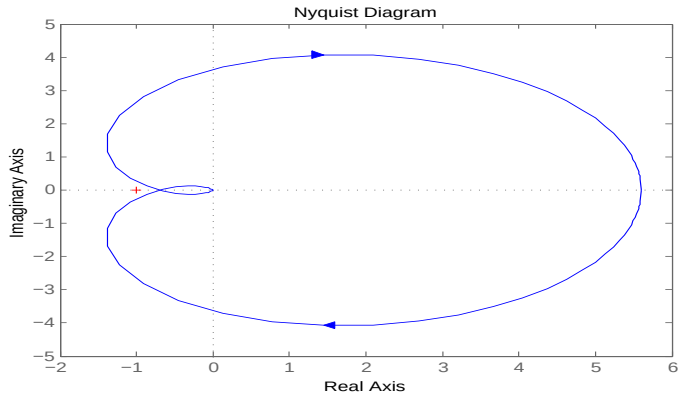
h) Now, $r(s) = 1/s^2$ and $e(s) = \frac{r(s)}{1+KG(s)}$. Therefore:

$$\boxed{\lim_{t\to\infty} e(t) = \lim_{s\to 0} se(s) = \lim_{s\to 0} \frac{sr(s)}{1+KG(s)} = \lim_{s\to 0} \frac{s+1}{s^2+s+K} = \frac{1}{K}.} \qquad [5]$$


Nyquist Diagram
Imaginary Axis
Real Axis
20
15
10
5
0
−5
−10
−15
−20
−1
−0.9
−0.8
−0.7
−0.6
−0.5
−0.4
−0.3
−0.2
−0.1
0

2. The transfer function used in fact was $G(s) = \frac{0.7}{(s+0.5)^3}$, although this is not required.

a) The intercepts can be obtained from the frequency response and are approximately given by $\boxed{5.6,\ -0.7 \text{ and } 0.}$ The Nyquist plot is shown below. [5]



b) The gain margin is approximately $\boxed{1.4}$ and the phase margin is approximately $\boxed{13^\circ}$. The cross-over frequency is approximately $\boxed{0.75 rads^{-1}}$. [5]

c) Let $K(s) = k$. The Nyquist criterion: $N = Z - P$, where N is the number of clockwise encirclements by the Nyquist diagram of $-k^{-1}$, P is the number of unstable open–loop poles and Z is the number of unstable closed–loop poles. Since $G(s)$ is stable, $P = 0$. Therefore: (a) when $k = 0.5$, $N = 0$ so $\boxed{Z = 0}$ (b) when $k = 5$, $N = 2$ so $\boxed{Z = 2}$ (c) when $k = -0.5$, $N = 1$ so $\boxed{Z = 1}$. [5]

d) $\boxed{K_1(s) \text{ is a PI compensator: } K_1(s) = K_P + \frac{K_I}{s} = \frac{K_I}{s}(1 + \frac{s}{K_I/K_P}).}$ It has high gain below $\omega_0 = K_I/K_P$ and gain close to K_P beyond ω_0. The phase is negative and large below ω_0 but insignificant above. So, by varying K_I and K_P we can use PI compensation to increase low frequency gain (hence improving tracking) without introducing phase-lag at high frequency (which reduces the phase margin) by placing w_0 in the 'middle' frequency range. Since the cross–over frequency for $G(s)$ is approximately 0.75 and ω_0 for $K_1(s)$ is approximately 0.1, this condition is satisfied. [5]

e) $\boxed{K_2(s) \text{ is a PD compensator: } K_2(s) = K_P + K_D s = K_D(s + K_P/K_D).}$ It has high gain above $\omega_0 = K_P/K_D$ and gain close to K_P below ω_0. The phase is positive and large above ω_0 but insignificant below. So, by varying K_D and K_P we can use PD compensation to increase the phase around the cross-over frequency gain (hence improving the phase margin) without introducing high gain at high frequency (which deteriorates the gain margin) by placing w_0 in the cross-over frequency range. Since the cross–over frequency for $G(s)$ is approximately 0.75 and ω_0 for $K_2(s)$ is approximately 1, this condition is satisfied. [5]

f) $\boxed{K_3(s) \text{ is a PID compensator: } K_3(s) = K_P + K_D s + \frac{K_I}{s}.}$ An inspection verifies that $K_3(s) = K_1(s)K_2(s)$ and so it combines the advantages of both PI and PD compensators. [5]

3. a) For a maximum overshoot of 5% and a settling time of 2 seconds the dominant closed–loop poles must be placed at $\boxed{s_1,\ \bar{s}_1 = -2 \pm j2.}$ [5]

b) i) The closed–loop poles are the roots of $1 + kG(s) = 0$ or $s^3 + 6s^2 + 12s + 8 + k = 0$. The Routh array is given by

$$\begin{array}{c|cc} s^3 & 1 & 12 \\ s^2 & 6 & 8+k \\ s & \frac{64-k}{6} & \\ 1 & 8+k & \end{array}$$

Thus for closed–loop stability, $\boxed{-8 < k < 64.}$ [5]

ii) The closed-loop characteristic equation is $1 + kG(s) = 0$ or

$$(s+2)^3 + k = s^3 + 6s^2 + 12s + 8 + k = 0.$$ [5]

iii) To achieve the design specifications, $s_1 = -2 + j2$ must satisfy the closed-loop characteristic equation for some real k. Equivalently, $(s_1 + 2)^3 + k = 0$ for some real k. However $\boxed{(s_1+2)^3 = -j8 \text{ is not real}}$, so the design specifications cannot be satisfied for any real k. [5]

c) A PD compensator has the form $K(s) = K_P + sK_D = K_D(s+z)$, $z = K_P/K_D$.

i) Taking the hint, we use z to cancel one of the poles of $G(s)$ so $z = 2$. The closed-loop characteristic equation in terms of K_D becomes $(s+2)^2 + K_D = 0$. Since this must be satisfied by $s_1 = -2 + j2$, $K_D = 4$. It follows that $K_P = 8$. So, $\boxed{K(s) = 4(s+2).}$ [5]

ii) Since $r(t)$ is a unit step, $r(s) = 1/s$. Let $e(s) = r(s) - y(s)$. Then

$$e(s) = \frac{1/s}{1 + 4/(s+2)^2}$$

The final value theorem then gives

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} se(s) = \lim_{s\to 0} \frac{1}{1 + 4/(s+2)^2}.$$

Therefore $\boxed{e_{ss} = 0.5.}$ [5]